Certificate-Based VPN Authentication for Cisco ISE with Intune (SCEP/PKCS)

Implementation Solution Guide

Version 1.0.0 · 2026-07-01

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Executive Summary — Certificate-Based VPN Authentication for Cisco ISE with Intune

The problem

Password-based VPN is the weak link Cisco ISE was built to remove. Passwords are phishable, shared, and reused; they give no proof that the connecting device is trusted. The fix — issuing the right device or user certificate to every endpoint, with a trust chain ISE will validate — defeats most teams because it spans three moving parts (a certificate authority, Intune delivery, and the Cisco head-end) that must line up exactly.

The solution

This package stands up **certificate-based VPN authentication end to end**. You issue a device or user certificate from **Microsoft Intune** — via **SCEP** (Certificate Connector / NDES or a Cloud PKI) or **PKCS** — deploy the **trusted-root** and certificate profiles so the chain is complete on each device, and hand **Cisco ISE** a clean **EAP-TLS identity** to authenticate. The result is passwordless VPN: the endpoint proves itself with a certificate instead of a secret.

It is delivered **pilot-first and rollback-ready**, following a disciplined **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** model with explicit pass/fail gates — cert present, chain valid, ISE EAP-TLS authentication success — so a misconfiguration is caught on a handful of pilot devices, never on the fleet.

What you get

- A **Solution Overview** explaining EAP-TLS with ISE as the RADIUS authority, the SCEP-vs-PKCS choice, and the trust chain ISE must validate.
- A **Requirements Guide** covering every prerequisite with how to verify and implement each.
- A click-by-click **Implementation Guide** for the Intune/Entra/certificate side.
- **Five PowerShell scripts** for readiness discovery, per-device certificate + chain validation, fleet coverage/expiry reporting, safe remediation, and pilot-group targeting.
- **Testing, Troubleshooting, and Best-Practices** guides with real pass/fail gates and field-proven renewal, revocation, and key-protection guidance.

Scope and boundary

The **Intune, Entra, and certificate** side is authored at exact, click-by-click detail. **Cisco ISE and VPN head-end console internals** (trust-store import, Certificate Authentication Profile, policy sets, identity sources, RADIUS/VPN connection profiles) are **deferred to Cisco's official documentation** — this solution guarantees ISE receives a correct, complete, chain-valid certificate to authenticate.

The bottom line

Every VPN user authenticated by a certificate Cisco ISE trusts — no passwords, delivered from Intune, and proven on a pilot before you scale. Self-guided and tested, at a fraction of the cost of a consulting engagement.

1. Solution Overview — Certificate-Based VPN Authentication for Cisco ISE with Intune (SCEP/PKCS)

What this solution does

This solution replaces **password-based VPN sign-in** with **certificate-based EAP-TLS authentication** that **Cisco ISE** validates. You issue a device or user certificate from **Microsoft Intune** — using either a **SCEP** or a **PKCS** certificate profile — deploy the **trusted-root** (and any intermediate) certificate so the chain is complete on the device, and present Cisco ISE with a clean **EAP-TLS identity** it can trust. The result is passwordless VPN authentication: the endpoint proves its identity with a certificate instead of a shared secret or a phishable password.

The work divides cleanly along a **vendor boundary**:

- **Intune / Entra / certificate side** — building the Trusted certificate profile, the SCEP or PKCS

certificate profile, the scope groups, assignment, and validation — is covered here at exact, click-by-click detail.

- **Cisco ISE side** — the policy set, the Certificate Authentication Profile, the identity source

sequence, and the RADIUS/VPN head-end configuration — is described here at the correct architectural level and **deferred to Cisco's official documentation** for exact console steps. This document does not fabricate ISE menu paths.

How EAP-TLS certificate VPN authentication works (with ISE as the RADIUS authority)

1. A VPN client (for example Cisco Secure Client / AnyConnect) initiates the VPN tunnel to the VPN

head-end (the ASA / Firepower / Secure Firewall).

1. The head-end acts as a **RADIUS client** and forwards authentication to **Cisco ISE**, which is

the **RADIUS authority (authentication server)**.

1. ISE and the endpoint perform an **EAP-TLS** handshake. In EAP-TLS the client presents its **client

certificate**; ISE validates that certificate against the** trust chain **it holds and the** Certificate Authentication Profile** rules you configure in ISE.

1. If the certificate is trusted, unexpired, not revoked, has the **Client Authentication** EKU, and

its identity attribute (Subject / SAN) matches an authorized identity, ISE returns **Access-Accept** and the VPN session is established — **no password is exchanged**.

Cisco ISE is the entity that must **trust and validate** the certificate. Intune is the entity that **delivers** the certificate and its trust chain to the endpoint. This solution makes those two sides meet correctly.

The trust chain ISE must validate

For ISE to accept an EAP-TLS certificate, the **entire chain** must line up:

- The endpoint holds a **leaf client certificate** issued by your issuing CA (via Intune SCEP/PKCS).
- The endpoint also holds the **root** (and any **intermediate**) certificate — delivered by the

Intune **Trusted certificate** profile — so the leaf chains to a trusted root **on the device**.

- **Cisco ISE** must independently hold and trust the **same root/intermediate** in its own

certificate trust store, so ISE can build and validate the chain when the endpoint presents its leaf. If ISE does not trust the issuing CA, EAP-TLS fails even when the certificate is perfectly installed on the device.

The chain must therefore be present in **two** places: on the **device** (Intune deploys it) and in **Cisco ISE** (you import it per Cisco's documentation). This is the single most common failure point and is called out again in Testing and Troubleshooting.

SCEP vs PKCS (which profile to use)

Intune can issue the client certificate two ways. Pick per your CA and identity model:

- **SCEP** — the device generates its **own private key** and requests a certificate using a

challenge issued through the **Intune Certificate Connector** (NDES/SCEP) or a **Cloud PKI** that supports SCEP. The private key **never leaves the device**. Best for **device** certificates and the broadest platform support (Windows, iOS/iPadOS, Android, macOS).

- **PKCS** — the certificate (and its private key) is issued **through the connector** against a CA

template and delivered to the device. Common for **user** certificates and scenarios that need a specific Subject/SAN pulled from **directory attributes** (for example the user's UPN in the SAN).

Guidance:

- Use **SCEP** for device-authenticated VPN where the machine identity is what ISE trusts, and where

keeping the private key on-device is a requirement.

- Use **PKCS** for user-authenticated VPN where ISE keys off the **user identity** (UPN/email in the

SAN) and you want the certificate subject to follow directory attributes.

- Many fleets run **one** of these for VPN; some run both (device cert for the machine tunnel, user

cert for the user tunnel). Decide the identity ISE will key on **first**, then choose the profile type that produces that identity.

Where Intune's certificate delivery meets the Cisco head-end

Layer	Owner	This solution
Issuing CA / Cloud PKI	Your CA / cloud PKI vendor	Named as a prerequisite; setup deferred to CA/vendor docs
Certificate delivery to endpoints (root + leaf)	**Intune**	**Full click-by-click detail here**

Layer	Owner	This solution
Scope / pilot targeting	**Intune + Entra groups**	**Full detail here**
Trust store on ISE (root/intermediate import)	**Cisco ISE**	Described architecturally; steps deferred to Cisco docs
EAP-TLS policy, Certificate Authentication Profile, identity source	**Cisco ISE**	Described architecturally; steps deferred to Cisco docs
VPN head-end / RADIUS client config	**Cisco ASA / Secure Firewall**	Described architecturally; steps deferred to Cisco docs

What you can control (Intune side)

- Which **trust chain** (root + intermediate) is deployed to devices, and to whom.
- The **Subject name** and **SAN** format of the issued certificate (from device/user attributes) — this is the identity ISE will read.
- **Key size / usage**, **validity period**, and **renewal threshold** (auto-renew before expiry).
- The **Client Authentication** EKU (`1.3.6.1.5.5.7.3.2`) — required for EAP-TLS.
- The **scope** of issuance via Entra security groups (pilot first, then expand).

Limitations (be honest with buyers)

- **Cisco ISE and the VPN head-end are out of scope for exact steps.** ISE trust-store import, the Certificate Authentication Profile, policy sets, identity source sequences, and ASA/Secure Firewall VPN configuration follow **Cisco's** documentation. This solution guarantees the Intune-issued certificate is correct, complete, and chain-valid — the input EAP-TLS needs.
- **You must import the same root/intermediate into ISE.** Deploying the chain to devices is not enough; ISE has its own trust store.
- **Identity mismatch is the classic failure.** The Subject/SAN Intune issues must match what your ISE Certificate Authentication Profile keys on (for example UPN in SAN). Design this together.
- **Revocation checking (CRL/OCSP) is enforced by ISE**, not Intune. If ISE is set to check revocation, the CA's CRL/OCSP endpoints must be reachable from ISE.

The standard deployment model

> ***Deploy the trusted chain → create the SCEP/PKCS certificate profile → issue to a named pilot group → confirm the cert + chain land on the device → validate ISE accepts the EAP-TLS handshake → expand to broad.*** *Pattern:* ***Assess → Configure → Pilot → Validate → Expand****, with an explicit pilot gate and a documented rollback.*

2. Requirements — Certificate-Based VPN Authentication for Cisco ISE with Intune

> *These are the **solution-specific** prerequisites. The **category-level** prerequisites (Entra tenant, Intune subscription, admin roles, device enrollment, pilot group, supported OS, network access, change control) are in* `global-requirements-security.md` *— confirm those first. Object-naming convention used throughout this solution (reuse verbatim): - Trusted root profile:* `Cisco VPN – Trusted Root` *- SCEP device certificate profile:* `Cisco VPN – SCEP Device Cert` *- SCEP user certificate profile:* `Cisco VPN – SCEP User Cert` *- PKCS user certificate profile:* `Cisco VPN – PKCS User Cert` *- Pilot scope group:* `PILOT – Cisco Cert VPN Users` *- Broad scope group:* `BROAD – Cisco Cert VPN Users`

Each requirement below is stated as **Description / Why required / How to verify / How to implement if missing.**

R1. An issuing Certificate Authority (on-prem CA or Cloud PKI)

Description: A CA that can issue **client-authentication** certificates — either an on-prem Microsoft Enterprise CA (AD CS) or a **Cloud PKI** service that integrates with Intune. **Why required:** The leaf certificate ISE validates must be issued by a CA. There is no certificate-based VPN without one. **How to verify:** For AD CS, the CA is online and has a certificate template with **Client Authentication** EKU. For Cloud PKI, the vendor's issuing CA is provisioned and its Intune integration is registered. **How to implement if missing:** Stand up an AD CS Enterprise CA and publish a client-auth template, **or** provision a Cloud PKI (for example Microsoft Cloud PKI or a third-party) per the vendor's documentation. CA build steps are deferred to Microsoft/CA-vendor docs.

R2. The Intune Certificate Connector (SCEP via NDES) — for on-prem CA SCEP/PKCS

Description: The **Intune Certificate Connector** installed on a Windows server, connected to your tenant, and (for SCEP) fronted by **NDES** with the SCEP/NDES policy module. PKCS also uses this connector. **Why required:** SCEP/PKCS issuance from an **on-prem** CA flows through this connector. (A Cloud PKI with native Intune SCEP may not need NDES — confirm with the vendor.) **How to verify: Intune admin center** → **Tenant administration** → **Connectors and tokens** → **Certificate connectors** (`https://intune.microsoft.com`) shows the connector with a recent **last-connection** time and an **Active** state. **How to implement if missing:** Install the Certificate Connector on a supported Windows Server, sign in with an Intune Administrator during setup, and (for SCEP) configure **NDES + the SCEP policy module**. Deferred to Microsoft's connector/NDES documentation.

R3. Trusted-root (and intermediate) certificate file(s)

Description: The **root CA certificate** (and any **intermediate** CA certificate) as `.cer` files, exported from your issuing CA / Cloud PKI. **Why required:** The Intune **Trusted certificate** profile deploys these so the device can chain the leaf certificate to a trusted root. Without the chain on the device, the leaf is unusable and EAP-TLS fails. **How to verify:** You have the root `.cer` (Base-64/DER) on hand; open it and confirm it is the CA that will issue the leaf. If there is an intermediate, you have that too. **How to implement if**

missing: Export the root/intermediate from the CA console (AD CS: **certutil -ca.cert** or the CA MMC) or download it from the Cloud PKI portal.

R4. Trusted-root certificate profile in Intune

Description: An Intune **Trusted certificate** profile named `Cisco VPN – Trusted Root` that deploys the root (and, if separate, a second profile or the same profile chain for the intermediate). **Why required:** It places the CA root/intermediate into the device's Trusted Root store so the leaf chains correctly. **How to verify: Intune admin center** → **Devices** → **Manage devices** → **Configuration** → **Policies**; the profile `Cisco VPN – Trusted Root` exists and is assigned. **How to implement if missing:** Build it in Implementation **Step 2** (this is core solution work).

R5. SCEP or PKCS certificate profile in Intune

Description: An Intune **SCEP** or **PKCS** certificate profile (named per convention, e.g. `Cisco VPN – SCEP Device Cert` or `Cisco VPN – PKCS User Cert`) that issues the leaf with the **Client Authentication** EKU and the Subject/SAN ISE will key on. **Why required:** This issues the actual certificate ISE authenticates. Its EKU and Subject/SAN must match what ISE expects. **How to verify:** The profile exists under **Configuration** → **Policies** and its EKU includes **Client Authentication**. **How to implement if missing:** Build it in Implementation **Step 3** (core solution work).

R6. Entra security groups for scope (pilot and broad)

Description: Two Entra security groups: `PILOT – Cisco Cert VPN Users` and `BROAD – Cisco Cert VPN Users` (device- or user-based to match your chosen certificate type). **Why required:** Certificate rollout is piloted before fleet-wide — mandatory for anything that touches authentication (see global R8, change control). **How to verify: Entra admin center** → **Groups** → **All groups** (`https://entra.microsoft.com`); both groups exist. **How to implement if missing:** Create them in Entra, or run `CISECERT-Set-ScopeTargeting.ps1` (included) to create them idempotently.

R7. Cisco ISE configured for EAP-TLS (trust store + certificate authentication)

Description: Cisco ISE is deployed as the RADIUS authority and configured to accept **EAP-TLS**: the **same root/intermediate** imported into ISE's trusted-certificate store, a **Certificate Authentication Profile** that reads the identity from the certificate (Subject or SAN), and a policy set that authorizes VPN access. **Why required:** ISE is the entity that trusts and validates the certificate. If ISE does not trust the issuing CA or is not configured for EAP-TLS, authentication fails regardless of a perfect device-side certificate. **How to verify:** In the ISE console, the issuing CA's root/intermediate appears in the trusted certificates store, a Certificate Authentication Profile exists, and EAP-TLS is allowed in the relevant policy set. **Exact ISE navigation is deferred to Cisco's documentation. How to implement if missing:** Import the CA chain into ISE, create the Certificate Authentication Profile, and enable EAP-TLS in the VPN policy set **per Cisco's official ISE documentation.** This solution provides the correct certificate for ISE to consume; the ISE console steps are Cisco's.

R8. Cisco VPN head-end pointing at ISE for RADIUS (EAP-TLS enabled)

Description: The VPN head-end (Cisco ASA / Secure Firewall) configured to use **ISE as its RADIUS server** for the VPN connection profile/tunnel-group, with EAP-TLS / certificate authentication enabled and a matching VPN client profile (e.g. Cisco Secure Client / AnyConnect). **Why required:** The head-end is the RADIUS client that hands the EAP-TLS session to ISE; without it the certificate is never presented to ISE. **How to verify:** The head-end's VPN connection profile references ISE as RADIUS and permits certificate/EAP-TLS authentication. **Exact head-end steps are deferred to Cisco's documentation. How to implement if missing:** Configure the ASA/Secure Firewall VPN connection profile to use ISE RADIUS with EAP-TLS **per Cisco's documentation.**

R9. Revocation reachability (if ISE enforces CRL/OCSP)

Description: If ISE is configured to check certificate revocation, the CA's **CRL distribution point** and/or **OCSP responder** must be reachable **from ISE**. **Why required:** ISE will reject an otherwise-valid certificate if it cannot complete a required revocation check. **How to verify:** ISE can reach the CRL/OCSP URLs embedded in the issued certificate (test from ISE / the network per Cisco docs). **How to implement if missing:** Publish the CRL/OCSP endpoints and allow ISE to reach them, or adjust ISE's revocation policy per Cisco's documentation.

3. Implementation — Certificate-Based VPN Authentication for Cisco ISE with Intune

This is the ordered, click-by-click procedure. Follow the steps in order — the **trust chain must be deployed before the certificate profile**, and everything is **piloted before it is expanded**. Deployment model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Object names used here (reuse verbatim): `Cisco VPN – Trusted Root` · `Cisco VPN – SCEP Device Cert` · `Cisco VPN – SCEP User Cert` · `Cisco VPN – PKCS User Cert` · `PILOT – Cisco Cert VPN Users` · `BROAD – Cisco Cert VPN Users`

Vendor boundary: Steps 1–6 (Intune/Entra/certificate) are exact. Step 7 (Cisco ISE + head-end) is described at the architectural level and points to Cisco's documentation for exact console steps — do not expect ISE menu paths here.

Step 0 — Assess and decide (before you touch Intune)

1. Confirm all requirements in `02-requirements.md` and the category `global-requirements-security.md`.
1. **Decide the identity ISE will key on.** This drives everything:

• **Device identity** (machine authenticates the tunnel) → use a **SCEP device** certificate (`Cisco VPN – SCEP Device Cert`). Typical Subject/SAN: device name / hardware identifier.

• **User identity** (user authenticates) → use a **SCEP user** or **PKCS user** certificate (`Cisco VPN – SCEP User Cert` / `Cisco VPN – PKCS User Cert`). Typical SAN: **UPN**.

1. Choose **SCEP vs PKCS** (see `01-solution-overview.md`). Default to **SCEP** unless you need a

directory-attribute Subject/SAN that your connector only produces via **PKCS**.

1. **Record the exact Subject/SAN format** you will issue — you must hand ISE the same identity to key on (coordinate with whoever configures ISE).

1. Ensure a change ticket exists with an approver, a maintenance window, and a rollback step (global R8).

Gate 0 (pass/fail): Identity model, profile type (SCEP/PKCS), and Subject/SAN format are decided and written down. **Fail** = stop; do not build profiles until the identity is agreed with ISE.

Step 1 — Create the scope groups

Create the pilot and broad groups first so profiles can be assigned to them.

Option A — portal:

1. Go to `https://entra.microsoft.com` → **Groups** → **All groups** → **New group**.
2. **Group type:** Security. **Group name:** `PILOT – Cisco Cert VPN Users`. Create.
3. Repeat for `BROAD – Cisco Cert VPN Users`.
4. Add a few **pilot** members (the test users, or their devices if you chose device certs) to `PILOT – Cisco Cert VPN Users`. Leave `BROAD` empty for now.

Option B — script: run `CISECERT-Set-ScopeTargeting.ps1 -Apply` to create both groups idempotently, then add pilot members.

Gate 1: Both groups exist; pilot group has test members.

Step 2 — Build the Trusted certificate profile (`Cisco VPN – Trusted Root`)

The chain must be on the device **before** the leaf.

1. In `https://intune.microsoft.com`, go to **Devices → Manage devices → Configuration → Policies → Create → New Policy**.

1. **Platform:** choose your target platform (e.g. **Windows 10 and later**). **Profile type:** **Templates** → **Trusted certificate**. Click **Create**.

1. **Name:** `Cisco VPN – Trusted Root`. **Description:** "Root CA for Cisco EAP-TLS VPN — deploys the trust chain the leaf certificate must chain to." **Next.**

1. **Configuration settings: Certificate file** → upload the **root** CA `.cer` (from R3).

For Windows, set **Destination store** = **Computer certificate store – Root**.

1. **Next** through Scope tags (optional).
2. **Assignments:** add group `PILOT – Cisco Cert VPN Users` to **Included groups**. **Next.**

3. **Review + create.**

4. **If you have an intermediate CA:** repeat Steps 2.1–2.7 with the intermediate `.cer`, naming it `Cisco VPN – Trusted Root (Intermediate)` and destination store **Computer certificate store – Intermediate**. Assign the same pilot group.

> ***Do the equivalent import on Cisco ISE (Step 7):*** *ISE needs the* ***same*** *root/intermediate in its own trust store. Deploying to devices alone is not sufficient.*

Gate 2: `Cisco VPN – Trusted Root` (and intermediate, if any) exists and is assigned to `PILOT – Cisco Cert VPN Users`.

Step 3 — Build the SCEP or PKCS certificate profile

Do **3A** for SCEP **or 3B** for PKCS — matching the choice from Step 0.

3A — SCEP certificate profile (`Cisco VPN – SCEP Device Cert` or `… SCEP User Cert`)

1. **Devices** → **Manage devices** → **Configuration** → **Policies** → **Create** → **New Policy.**

2. **Platform:** your platform. **Profile type: Templates** → **SCEP certificate. Create.**

3. **Name:** `Cisco VPN – SCEP Device Cert` (device) **or** `Cisco VPN – SCEP User Cert` (user). **Next.**

1. **Configuration settings:**

• **Certificate type: Device** or **User** (match your identity decision).

• **Subject name format:** set to the identity you agreed with ISE (e.g. `CN={{DeviceName}}` for device, or `CN={{UserPrincipalName}}` for user).

• **Subject alternative name:** add the SAN ISE keys on — commonly **UPN** = `{{UserPrincipalName}}` for user certs. **This must match the ISE Certificate Authentication Profile.**

• **Certificate validity period:** e.g. **1 year**.

• **Key storage provider (KSP):** Enroll to TPM if present, otherwise Software KSP (Windows).

• **Key usage:** Digital signature + Key encipherment.

• **Extended key usage (EKU):** add **Client Authentication** (`1.3.6.1.5.5.7.3.2`). **Required for EAP-TLS.**

• **Renewal threshold (%):** e.g. **20%** (auto-renew before expiry).

• **Root Certificate:** select the `Cisco VPN – Trusted Root` profile from Step 2.

• **SCEP Server URLs:** the NDES/SCEP URL from your Certificate Connector (or the Cloud PKI SCEP URL).

1. **Next** → Scope tags (optional) → **Assignments:** include `PILOT – Cisco Cert VPN Users`.

2. **Review + create.**

3B — PKCS certificate profile (`Cisco VPN – PKCS User Cert`)

1. **Devices** → **Manage devices** → **Configuration** → **Policies** → **Create** → **New Policy.**
2. **Platform:** your platform. **Profile type: Templates** → **PKCS certificate. Create.**
3. **Name:** `Cisco VPN – PKCS User Cert`. **Next.**
4. **Configuration settings:**

• **Certification authority:** the FQDN of your issuing CA.

• **Certification authority name:** the CA common name.

• **Certificate template name:** the client-auth template published on the CA (R1).

• **Certificate type: User** (typical for PKCS).

• **Subject name format / SAN:** the identity ISE keys on (e.g. SAN **UPN** = `{{UserPrincipalName}}`).

• **Extended key usage:** include **Client Authentication** (`1.3.6.1.5.5.7.3.2`).

• **Root Certificate:** select `Cisco VPN – Trusted Root` from Step 2.

• **Renewal threshold (%):** e.g. **20%**.

1. **Next** → Scope tags (optional) → **Assignments:** include `PILOT – Cisco Cert VPN Users`.
2. **Review + create.**

Gate 3: The SCEP or PKCS profile exists, includes the **Client Authentication** EKU, references `Cisco VPN – Trusted Root`, issues the agreed Subject/SAN, and is assigned to the pilot group.

Step 4 — Confirm the certificate lands on a pilot device

1. On a pilot device (a member of `PILOT – Cisco Cert VPN Users`), force an Intune sync (**Settings** → **Accounts** → **Access work or school** → **Info** → **Sync**, or wait for check-in).
    1. Run `CISECERT-Validation.ps1 -IssuingCaName "your-issuing-ca-name"` on the device (elevated). It confirms: root trusted, leaf present and issued by your CA, chain valid, **Client Authentication** EKU present, and within validity.
    1. Alternatively check manually: `certlm.msc` → **Personal** → **Certificates** shows the leaf; **Trusted Root / Intermediate** shows the CA chain.

Gate 4 (device-side): Validation reports **PASS** on all checks — cert present, chain valid, EKU present. **Fail** = go to Troubleshooting **A (no cert)** / **B (chain incomplete)** before proceeding to ISE. **Do not test ISE until this gate passes.**

Step 5 — Validate ISE accepts the EAP-TLS handshake (vendor boundary)

This step happens on the **Cisco** side. It is described here architecturally; **exact ISE / head-end console steps are Cisco's** — follow Cisco's official documentation.

1. **Ensure ISE trusts the CA:** the same **root/intermediate** from Step 2 is imported into ISE's

trusted-certificate store. *(Cisco docs — ISE trusted certificates import.)*

1. **Ensure a Certificate Authentication Profile** in ISE reads the identity from the **same

Subject/SAN** you issued in Step 3 (e.g. SAN UPN). *(Cisco docs — Certificate Authentication Profile.)*

1. **Ensure the VPN policy set** allows **EAP-TLS** and authorizes the pilot identity for VPN.

(Cisco docs — policy sets / authorization.)

1. **Ensure the head-end** (ASA / Secure Firewall) points its VPN connection profile at ISE for

RADIUS with certificate/EAP-TLS auth. *(Cisco docs — VPN connection profile / RADIUS.)*

1. **Connect the VPN** from the pilot device using the certificate identity (no password).
2. **Confirm in ISE Live Logs** that the authentication is **EAP-TLS** and the result is

Access-Accept / success. *(Cisco docs — RADIUS Live Logs.)*

Gate 5 (end-to-end): ISE Live Logs show a successful **EAP-TLS Access-Accept** for the pilot identity and the VPN session establishes **without a password**. **Fail** = Troubleshooting **C (ISE rejects EAP-TLS)**.

Step 6 — Pilot soak

1. Keep the pilot group live for your agreed soak window (e.g. 3–5 business days).
2. Run `CISECERT-Report.ps1` to confirm the pilot devices show **Succeeded** on the certificate

profile and no `Error`/`Pending` laggards.

1. Watch for renewal behavior and any ISE rejections in Live Logs.

Gate 6: Pilot is stable for the full window — certs issued and renewing, ISE consistently accepting EAP-TLS, zero password fallbacks. **Fail** = fix before expanding.

Step 7 — Expand to broad

1. Add production members to `BROAD – Cisco Cert VPN Users` (or convert it to a dynamic group).
2. Add `BROAD – Cisco Cert VPN Users` to the **Included groups** of `Cisco VPN – Trusted Root`

(and intermediate) **and** the SCEP/PKCS profile — the **same profiles** you piloted.

1. Expand in waves; after each wave run `CISECERT-Report.ps1` to confirm coverage and chase

laggards, and spot-check ISE Live Logs.

1. Once broad coverage is confirmed and stable, plan removal of password-based VPN fallback per your

change process.

Gate 7: Target fleet shows certificate coverage, ISE accepts EAP-TLS across the fleet, and the report shows no unresolved `Error`/`Pending`. Rollout complete.

Rollback (any gate fails)

• **Unassign** the SCEP/PKCS profile (and Trusted Root if needed) from the affected group — this stops new issuance; existing tunnels using passwords remain available if you have not yet removed the password fallback.

• **Do not** delete issued certificates from devices reactively; unassigning the profile and letting ISE fall back to the prior VPN auth method is the safe path.

• Keep the password/legacy VPN auth method enabled on the head-end until broad certificate auth is proven (Gate 6+), so a failure never locks users out of VPN.

5. Testing & Validation — Certificate-Based VPN Authentication for Cisco ISE with Intune

Run these tests in order. Each has an explicit **pass/fail gate**. Do not advance past a failed gate — a failure here prevents a lockout in production. The three gates that matter most: **cert present, chain valid, ISE EAP-TLS auth success.**

> *Object names referenced:* `Cisco VPN – Trusted Root`, `Cisco VPN – SCEP Device Cert` / `Cisco VPN – SCEP User Cert` / `Cisco VPN – PKCS User Cert`, `PILOT – Cisco Cert VPN Users`.

Test 1 — Profiles exist and are assigned (readiness)

How: Run `CISECERT-Discovery.ps1`. Confirm the certificate connector is present/active and that `Cisco VPN – Trusted Root` and your SCEP/PKCS profile appear with the pilot group assigned. Alternatively, in **Intune** → **Devices** → **Manage devices** → **Configuration** → **Policies**, confirm both profiles and their assignments. **PASS:** Connector active; Trusted Root + SCEP/PKCS profiles exist and are assigned to `PILOT – Cisco Cert VPN Users`. **FAIL:** Missing connector or profile → Troubleshooting **A**; revisit Implementation Steps 2–3.

Test 2 — The trust chain is present on the device

How: On a pilot device, run `CISECERT-Validation.ps1 -IssuingCaName "your-issuing-ca-name"` (elevated). Check the **root trusted** result. Or open `certlm.msc` → **Trusted Root Certification Authorities** (and **Intermediate** if applicable) and confirm the CA is present. **PASS:** The root (and intermediate) CA certificate is present in the device trust store. **FAIL:** Chain incomplete → Troubleshooting **B**. Do not proceed — the leaf cannot chain.

Test 3 — The leaf certificate is issued and valid (cert present)

How: On the pilot device, from the same `CISECERT-Validation.ps1` run, check: **leaf present**, **issued by your CA**, **chain valid**, **Client Authentication EKU present**, **within validity**. Or `certlm.msc` → **Personal** → **Certificates** and inspect the leaf's **Issuer**, **EKU** (Client Authentication), and **valid-to** date. **PASS:** A leaf certificate issued by your CA is present, chains to the trusted root, carries the **Client Authentication** EKU (`1.3.6.1.5.5.7.3.2`), and is unexpired. **FAIL:** No cert / wrong EKU / broken chain → Troubleshooting **A** (no cert) or **B** (chain).

Test 4 — Subject/SAN matches what ISE keys on (identity)

How: Inspect the leaf's **Subject** and **Subject Alternative Name** (`certlm.msc` → cert → Details, or the Subject line printed by `CISECERT-Validation.ps1`). Compare to the identity attribute your ISE **Certificate Authentication Profile** reads (e.g. SAN **UPN**). **PASS:** The issued Subject/SAN exactly matches the attribute ISE is configured to key on. **FAIL:** Mismatch → fix the SCEP/PKCS **Subject name / SAN** (Implementation Step 3) or align the ISE Certificate Authentication Profile (Cisco docs). This is the classic silent failure.

Test 5 — ISE accepts the EAP-TLS handshake (auth success)

How: Connect the VPN from the pilot device using the certificate (no password). Confirm success in **ISE RADIUS Live Logs** (*Cisco docs*): the authentication method is **EAP-TLS** and the result is **Access-Accept**. **PASS:** ISE Live Logs show **EAP-TLS** → **Access-Accept** for the pilot identity and the VPN session establishes **without a password**. **FAIL:** ISE rejects → Troubleshooting **C**. Common causes: ISE does not trust the CA, the Certificate Authentication Profile keys on a different attribute, revocation check fails, or the head-end is not passing EAP-TLS.

Test 6 — Renewal works before expiry

How: Confirm the profile's **renewal threshold** is set (e.g. 20%). For a fast check, review a pilot device near threshold, force a sync, and re-run `CISECERT-Validation.ps1` to confirm a fresh `NotAfter`. Use `CISECERT-Report.ps1` to watch for devices trending toward expiry. **PASS:** Certificates renew automatically before expiry; no device drops to an expired leaf. **FAIL:** Renewal not firing → Troubleshooting **D** (renewal failures).

Test 7 — Fleet coverage before expansion

How: Run `CISECERT-Report.ps1` (optionally `-CsvPath`). Review Succeeded / Error / Pending counts for the certificate profile. **PASS:** Pilot (then each expansion wave) shows the expected Succeeded count with no unresolved Error/Pending. **FAIL:** Laggards present → run `CISECERT-Remediation.ps1` (default `-WhatIf`, then `-Apply`) to trigger re-evaluation, re-check, then expand.

Exit criteria (all must pass before broad rollout)

• Test 1 (readiness), Test 2 (**chain valid**), Test 3 (**cert present**), Test 4 (identity), Test 5 (**ISE EAP-TLS auth success**), Test 6 (renewal), Test 7 (coverage) all **PASS** on the pilot.

• A rollback step is documented and the password/legacy VPN auth path remains available until broad certificate auth is proven.

6. Troubleshooting — Certificate-Based VPN Authentication for Cisco ISE with Intune

Work top-down: confirm the **certificate is on the device (A)**, then the **chain is complete (B)**, then that **ISE accepts EAP-TLS (C)**, then **renewal (D)**. Most "ISE rejects" cases are actually a device-side chain or identity problem — fix A/B before touching ISE.

> *Diagnostic scripts:* `CISECERT-Discovery.ps1` *(tenant readiness),* `CISECERT-Validation.ps1` *(this device),* `CISECERT-Report.ps1` *(fleet),* `CISECERT-Remediation.ps1` *(safe re-trigger).*

A. No certificate on the device

Symptom: `CISECERT-Validation.ps1` reports no leaf in `LocalMachine\My` (or user store); the SCEP/PKCS profile shows **Error**/**Pending** in Intune.

Check and fix:

1. **Assignment.** Is the device/user actually in `PILOT – Cisco Cert VPN Users`, and is that group assigned to the SCEP/PKCS profile? (`CISECERT-Discovery.ps1` shows assigned groups.)

1. **Device check-in.** Force an Intune sync; issuance only happens on check-in. Devices offline will not receive the cert.

1. **Connector health.** In **Intune → Tenant administration → Connectors and tokens → Certificate connectors**, confirm the connector is** Active** with a recent last-connection. A stale/errored connector blocks all SCEP/PKCS issuance.

1. **SCEP URL / NDES.** For SCEP, verify the SCEP Server URL in the profile is correct and NDES is reachable/healthy. For PKCS, verify the CA name and template name are correct and the CA is online.

1. **Template permissions (PKCS/AD CS).** The connector's service account must have **Enroll** rights on the client-auth template.

1. **Re-trigger.** Run `CISECERT-Remediation.ps1 -CertProfileMatch "Cisco VPN" -Apply` to send a sync and re-evaluate issuance, then re-check with `CISECERT-Report.ps1`.

B. Chain incomplete / certificate not trusted on the device

Symptom: The leaf is present but `CISECERT-Validation.ps1` reports **chain valid = FAIL**, or the root is not found in the Trusted Root store.

Check and fix:

1. **Trusted Root profile assigned?** Confirm `Cisco VPN – Trusted Root` is assigned to the same group and has reached the device (`certlm.msc → Trusted Root`).
1. **Missing intermediate.** If your CA has an intermediate, you need `Cisco VPN – Trusted Root (Intermediate)` deployed to the **Intermediate** store. A missing intermediate is the most common broken-chain cause.
1. **Right store.** Windows device certs must chain via the **Computer** (LocalMachine) stores; a root deployed to the user store won't help a machine cert.
1. **Correct root.** Verify the deployed root is the CA that actually **issued** the leaf (issuer of the leaf == subject of the root).
1. Re-run `CISECERT-Validation.ps1` until **chain valid = PASS** before testing ISE.

C. ISE rejects the EAP-TLS handshake

Symptom: VPN connects with a certificate but fails; ISE Live Logs show EAP-TLS failure / Access-Reject (or no EAP-TLS at all). *(ISE console steps are Cisco's — this section tells you what to check and where the boundary is.)*

Device-side first (fix before blaming ISE):

1. **Chain valid on device?** Confirm Test 2/3 PASS — a device that can't build its own chain will fail EAP-TLS.
1. **Client Authentication EKU present?** `CISECERT-Validation.ps1` reports this. Missing EKU → fix the SCEP/PKCS profile EKU (Implementation Step 3) and reissue.
1. **Identity match (the classic).** The leaf **Subject/SAN** must match the attribute the ISE **Certificate Authentication Profile** keys on (Test 4). A UPN-in-SAN cert against an ISE profile keyed on Subject CN silently fails.

ISE-side (per Cisco documentation):

1. **ISE trusts the CA.** The same **root/intermediate** must be imported into ISE's trusted-certificate store. If ISE doesn't trust the issuing CA, every EAP-TLS attempt is rejected. *(Cisco docs — ISE trusted certificates.)*
1. **EAP-TLS allowed in the policy set** and the identity authorized for VPN. *(Cisco docs — policy sets / authorization.)*
1. **Revocation reachability.** If ISE enforces CRL/OCSP, ISE must reach the CA's CRL/OCSP endpoints

(R9); an unreachable CRL causes rejects even for valid certs. *(Cisco docs — revocation settings.)*

1. **Head-end passing EAP-TLS.** The ASA/Secure Firewall VPN connection profile must forward certificate/EAP-TLS auth to ISE as RADIUS. *(Cisco docs — VPN connection profile / RADIUS.)*

1. **Read the ISE Live Log detail** — it names the exact failure reason (unknown CA, identity not found, revocation, expired), which maps back to one of the checks above.

D. Renewal failures

Symptom: Certificates expire instead of renewing; users lose VPN at cert expiry; `CISECERT-Report.ps1` shows devices trending to expiry or dropping to Error.

Check and fix:

1. **Renewal threshold set?** Confirm the SCEP/PKCS profile has a **renewal threshold (%)** (e.g. 20%). Without it, no auto-renewal.

1. **Device checking in?** Renewal only fires on Intune check-in; offline/stale devices miss the window. Chase stale check-ins.

1. **Connector/CA still healthy?** Renewal re-requests through the same connector/CA path as initial issuance — re-run Section A checks 3–5.

1. **Same identity on renewal.** Confirm the renewed cert keeps the **same Subject/SAN** so ISE still accepts it; a changed SAN format breaks a previously working device.

1. **ISE revocation of the old cert.** Renewal issues a new cert; ensure the superseded cert isn't still being presented (clear old certs if a device accumulates multiples with the same EKU).

Quick reference: symptom → section

Symptom	Go to
No leaf certificate on device; profile Error/Pending	A
Leaf present but chain invalid / root not trusted	B
VPN cert connects but ISE rejects / Access-Reject	C
Cert expires instead of renewing	D
Not sure if profiles/connector are even set up	Run `CISECERT-Discovery.ps1`
Need fleet-wide coverage/expiry picture	Run `CISECERT-Report.ps1`

7. Best Practices — Certificate-Based VPN Authentication for Cisco ISE with Intune

Field-proven practices for running certificate-based EAP-TLS VPN authentication safely over time. Focus areas: **renewal, revocation, and key protection.**

Renewal

• **Set a renewal threshold** on every SCEP/PKCS profile (e.g. **20%** of validity). Auto-renewal before expiry is what keeps users from silently losing VPN access.

• **Keep validity periods sane.** Shorter validity (e.g. 1 year) limits exposure but means renewal must be reliable; do not set validity so short that renewal churn becomes a support burden.

• **Monitor upcoming expiry** with `CISECERT-Report.ps1` on a schedule; chase devices that miss a renewal window before their leaf expires.

• **Confirm the device checks in.** Renewal only fires when the device syncs with Intune — devices offline past their threshold will not renew. Track stale check-ins.

• **Test renewal in the pilot**, not in production. Verify a device near threshold actually receives a fresh certificate and that ISE accepts the renewed cert (same Subject/SAN).

Revocation

• **Decide who enforces revocation.** In this design **Cisco ISE** performs revocation checking (CRL/OCSP) during EAP-TLS. Make that policy explicit with whoever owns ISE.

• **Ensure CRL/OCSP reachability from ISE** (R9). A valid certificate is rejected if ISE cannot complete a required revocation check — keep the CA's CRL distribution point / OCSP responder published and reachable.

• **Have a revocation runbook.** When a device is lost/retired, revoke its certificate at the CA (or via the Cloud PKI / connector) **and** retire the device in Intune. Revocation on the CA is what makes ISE deny a compromised certificate; removing the Intune assignment alone does not revoke an already-issued cert.

• **Keep CRL publication frequent enough** that revocation takes effect within your risk tolerance; prefer **OCSP** where low-latency revocation matters.

Key protection

• **Prefer SCEP for device certs** so the **private key is generated on the device and never leaves it**.

• **Use the TPM** as the key storage provider where available (Windows) so the private key is hardware-protected and non-exportable.

• **Mark keys non-exportable.** Do not issue exportable private keys for VPN authentication unless a specific, documented reason exists.

• **For PKCS**, understand the key is generated at the connector/CA and delivered — protect the connector server accordingly and restrict who can issue against the client-auth template.

Identity and trust hygiene

• **One agreed identity attribute.** Fix the Subject/SAN Intune issues and the attribute ISE keys on as a single documented decision; changing one without the other silently breaks authentication.

• **Import the chain into ISE deliberately.** Track which root/intermediate ISE trusts; rotate ISE's trust when the CA rotates. Deploying a new chain to devices without updating ISE breaks EAP-TLS.

• **Least-privilege EKU.** Issue leaf certificates with **Client Authentication** only unless another EKU is genuinely required.

Rollout and operations

• **Always pilot first**, with a documented rollback and the password/legacy VPN path kept available until certificate auth is proven (Gate 6+). Never remove the fallback before broad success.

• **Expand in waves**, validating coverage with `CISECERT-Report.ps1` and spot-checking ISE Live Logs each wave.

• **Name every object per convention** (`Cisco VPN – …`, `PILOT – …`, `BROAD – …`) so the Intune, Entra, and ISE sides stay legible to the next operator.

• **Keep change control** on anything touching certificates or VPN auth (global R8) — these changes can lock users out of the network.

• **Respect the vendor boundary.** Own the Intune/Entra/certificate side precisely; follow Cisco's official documentation for ISE and head-end configuration and keep those steps in your ISE runbook.

8. Deliverables Summary — Certificate-Based VPN Authentication for Cisco ISE with Intune

What's in this solution

A complete, self-guided package to replace password-based Cisco VPN sign-in with **certificate-based EAP-TLS authentication** — certificates issued from **Intune** (SCEP or PKCS) with a trust chain **Cisco ISE** validates, deployed pilot-first and rollback-ready.

Documents

File	What it gives you
`01-solution-overview.md`	How EAP-TLS certificate VPN auth works with ISE as the RADIUS authority; SCEP vs PKCS; the trust chain ISE must validate; where Intune delivery meets the Cisco head-end.
`02-requirements.md`	Every prerequisite (CA/Cloud PKI, Certificate Connector/NDES, trusted-root + SCEP/PKCS profiles, Entra groups, ISE for EAP-TLS, head-end, revocation) with Description / Why / Verify / Implement.
`03-implementation.md`	Ordered, click-by-click steps: scope groups → Trusted Root profile → SCEP/PKCS profile → confirm cert lands → validate ISE EAP-TLS → pilot → expand, with pass/fail gates and rollback.
`05-testing.md`	Seven validation tests with explicit gates (cert present, chain valid, ISE auth success).
`06-troubleshooting.md`	The failure modes: no cert, chain incomplete, ISE rejects EAP-TLS, renewal failures — with concrete fixes.
`07-best-practices.md`	Renewal, revocation, and key-protection practices for steady-state operation.
`global-requirements-security.md`	Category-level prerequisites shared across all security solutions.
`executive-summary.md`	One-page business case for decision-makers.

PowerShell scripts (read-only / safe by default)

Script	Purpose	Default posture
`CISECERT-Discovery.ps1`	Certificate readiness: Intune certificate connectors + trusted-root/SCEP/PKCS profiles (Graph).	Read-only
`CISECERT-Validation.ps1`	Confirm the VPN cert + full chain are present on **this** device (local cert store checks).	Read-only
`CISECERT-Report.ps1`	Certificate deployment + expiry coverage across the fleet (Graph).	Read-only
`CISECERT-Remediation.ps1`	Safely re-trigger missing/failed cert issuance.	`-WhatIf` by default
`CISECERT-Set-ScopeTargeting.ps1`	Create/populate the pilot/broad Entra group(s).	`-WhatIf` by default

All Graph scripts require the Microsoft Graph PowerShell SDK and read-only scopes (Remediation and group creation request write scopes only for their explicit `-Apply` action).

Named objects you will create (reuse verbatim)

- `Cisco VPN – Trusted Root` (and `Cisco VPN – Trusted Root (Intermediate)` if applicable)
- `Cisco VPN – SCEP Device Cert` / `Cisco VPN – SCEP User Cert` / `Cisco VPN – PKCS User Cert`
- `PILOT – Cisco Cert VPN Users` / `BROAD – Cisco Cert VPN Users`

Vendor boundary

The Intune / Entra / certificate side is documented at exact, click-by-click detail. **Cisco ISE and VPN head-end (ASA / Secure Firewall) console configuration — trust-store import, Certificate Authentication Profile, policy sets, identity sources, RADIUS/VPN connection profiles — is deferred to Cisco's official documentation.** This solution guarantees ISE receives a correct, complete, chain-valid EAP-TLS certificate to authenticate.

Definition of done

Every targeted VPN user authenticated by a certificate Cisco ISE trusts — no passwords, issued from Intune, proven on a pilot (Gates 0–6) before expanding to the fleet (Gate 7), with renewal and revocation operationalized and a rollback path kept available throughout.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

CISECERT-Discovery.ps1

Purpose: Enumerates the tenant's Intune certificate connectors (state / last check-in) and the existing certificate device configuration profiles (Trusted certificate, SCEP, PKCS) with their assigned groups, so you can confirm the certificate infrastructure for Cisco EAP-TLS VPN is in place before you issue certificates. Read-only against Microsoft Graph. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\CISECERT-Discovery.ps1 Requires the Microsoft Graph PowerShell SDK and consent to: DeviceManagementConfiguration.Read.All, DeviceManagementServiceConfig.Read.All. EXPECTED RESULTS A list of certificate connectors with state / last-connection, and a list of trusted-cert / SCEP / PKCS profiles with their assigned group IDs. Flags if no connector or profile is found. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. Cisco ISE internals deferred to vendor docs. #> [CmdletBinding()] param()

How to run: powershell -ExecutionPolicy Bypass -File .\CISECERT-Discovery.ps1 Requires the Microsoft Graph PowerShell SDK and consent to: DeviceManagementConfiguration.Read.All, DeviceManagementServiceConfig.Read.All. EXPECTED RESULTS A list of certificate connectors with state / last-connection, and a list of trusted-cert / SCEP / PKCS profiles with their assigned group IDs. Flags if no connector or profile is found. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. Cisco ISE internals deferred to vendor docs. #> [CmdletBinding()] param()

CISECERT-Remediation.ps1

Purpose: For devices whose Cisco VPN SCEP/PKCS certificate profile shows Error/Pending, performs the safe, reversible remediation that resolves the common causes: triggers a device sync/check-in so the certificate profile re-evaluates and re-requests issuance. It does NOT delete certificates, unassign profiles, or touch any CA. By default it only REPORTS what it would do (-WhatIf); pass -Apply to act. HOW TO RUN Report only (default): powershell -ExecutionPolicy Bypass -File .\CISECERT-Remediation.ps1 -CertProfileMatch "Cisco VPN" Apply: powershell -ExecutionPolicy Bypass -File .\CISECERT-Remediation.ps1 -CertProfileMatch "Cisco VPN" -Apply Requires the Microsoft Graph PowerShell SDK and consent to: DeviceManagementConfiguration.Read.All, DeviceManagementManagedDevices.Read.All, DeviceManagementManagedDevices.PrivilegedOperations.All (to send a sync action). EXPECTED RESULTS A list of devices in Error/Pending on the matched Cisco VPN cert profiles; with -Apply, a sync action is issued to each so issuance re-evaluates at next check-in. Re-run CISECERT-Report.ps1 after a sync cycle to confirm the move to Succeeded. .NOTES White-label — the repeatable way. Safe by default (-WhatIf). Connector/CA repair is out of scope; Cisco ISE internals deferred to vendor docs. #> [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='Medium')] param([string]$CertProfileMatch = 'Cisco VPN', [switch]$Apply)

How to run: Report only (default): powershell -ExecutionPolicy Bypass -File .\CISECERT-Remediation.ps1 -CertProfileMatch "Cisco VPN" Apply: powershell -ExecutionPolicy Bypass -File .\CISECERT-Remediation.ps1 -CertProfileMatch "Cisco VPN" -Apply Requires the Microsoft Graph PowerShell SDK and consent to: DeviceManagementConfiguration.Read.All, DeviceManagementManagedDevices.Read.All, DeviceManagementManagedDevices.PrivilegedOperations.All (to send a sync action). EXPECTED

RESULTS A list of devices in Error/Pending on the matched Cisco VPN cert profiles; with -Apply, a sync action is issued to each so issuance re-evaluates at next check-in. Re-run CISECERT-Report.ps1 after a sync cycle to confirm the move to Succeeded. .NOTES White-label — the repeatable way. Safe by default (-WhatIf). Connector/CA repair is out of scope; Cisco ISE internals deferred to vendor docs. #> [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='Medium')] param([string]$CertProfileMatch = 'Cisco VPN', [switch]$Apply)

CISECERT-Report.ps1

Purpose: Builds a fleet-wide picture of certificate coverage to drive the pilot and expansion gates: for each managed device it reports the SCEP/PKCS certificate profile deployment state (Succeeded / Error / Pending) for the Cisco VPN certificate profiles, so coverage and laggards are visible before you expand. Read-only against Microsoft Graph. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\CISECERT-Report.ps1 -CertProfileMatch "Cisco VPN" -CsvPath .\cisecert-coverage.csv Requires the Microsoft Graph PowerShell SDK and consent to: DeviceManagementConfiguration.Read.All, DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS A per-profile deployment-state summary (how many devices Succeeded / Error / Pending) for the matched Cisco VPN cert profiles, with an optional CSV export. Exact per-device certificate expiry (NotAfter) lives on the device — run CISECERT-Validation.ps1 there for precise dates. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. Cisco ISE internals deferred to vendor docs. #> [CmdletBinding()] param([string]$CertProfileMatch = 'Cisco VPN', [string]$CsvPath)

How to run: powershell -ExecutionPolicy Bypass -File .\CISECERT-Report.ps1 -CertProfileMatch "Cisco VPN" -CsvPath .\cisecert-coverage.csv Requires the Microsoft Graph PowerShell SDK and consent to: DeviceManagementConfiguration.Read.All, DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS A per-profile deployment-state summary (how many devices Succeeded / Error / Pending) for the matched Cisco VPN cert profiles, with an optional CSV export. Exact per-device certificate expiry (NotAfter) lives on the device — run CISECERT-Validation.ps1 there for precise dates. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. Cisco ISE internals deferred to vendor docs. #> [CmdletBinding()] param([string]$CertProfileMatch = 'Cisco VPN', [string]$CsvPath)

CISECERT-Set-ScopeTargeting.ps1

Purpose: Creates (idempotently) the Entra security groups used to scope Cisco EAP-TLS VPN certificate issuance, following the solution naming convention so the Intune Trusted Root and SCEP/PKCS profiles can be assigned consistently: PILOT - Cisco Cert VPN Users BROAD - Cisco Cert VPN Users Existing groups (matched by displayName) are reused, not duplicated. Members are NOT added — add pilot users/devices by hand or via dynamic rules per your standard. HOW TO RUN Report only (default): powershell -ExecutionPolicy Bypass -File .\CISECERT-Set-ScopeTargeting.ps1 Apply: powershell -ExecutionPolicy Bypass -File .\CISECERT-Set-ScopeTargeting.ps1 -Apply Requires the Microsoft Graph PowerShell SDK and consent to: Group.ReadWrite.All. EXPECTED RESULTS The two scope groups exist (created if missing, reused if present), with their object IDs printed for use in the Intune profile assignments (Implementation Steps 2-3). .NOTES White-label — the repeatable way. Read-only unless -Apply (safe by default -WhatIf). Cisco ISE internals deferred to vendor docs. #> [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='Medium')] param([switch]$Apply)

How to run: Report only (default): powershell -ExecutionPolicy Bypass -File .\CISECERT-Set-ScopeTargeting.ps1 Apply: powershell -ExecutionPolicy Bypass -File .\CISECERT-Set-ScopeTargeting.ps1 -Apply Requires the Microsoft Graph PowerShell SDK and consent

to: Group.ReadWrite.All. EXPECTED RESULTS The two scope groups exist (created if missing, reused if present), with their object IDs printed for use in the Intune profile assignments (Implementation Steps 2-3). .NOTES White-label — the repeatable way. Read-only unless -Apply (safe by default -WhatIf). Cisco ISE internals deferred to vendor docs. #> [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='Medium')] param([switch]$Apply)

CISECERT-Validation.ps1

Purpose: Run on a pilot/managed Windows device to confirm the EAP-TLS VPN certificate is actually deliverable: the issuing CA root/intermediate is present and trusted, a leaf client certificate was issued by that CA, chains to the trusted root, has the Client Authentication EKU, and is within validity. This is the device-side proof behind Testing Tests 2-4. Read-only. HOW TO RUN Elevated on the device: powershell -ExecutionPolicy Bypass -File .\CISECERT-Validation.ps1 -IssuingCaName "your-issuing-ca-name" -IssuingCaName is matched against certificate Issuer/Subject to identify the VPN certificate chain. EXPECTED RESULTS PASS for: issuing CA root trusted; a leaf cert issued by that CA present in LocalMachine\My; chain valid; Client Authentication EKU present; not expired; plus the Subject/SAN printed so you can confirm it matches what Cisco ISE keys on. Any FAIL maps to Troubleshooting A/B. .NOTES White-label — the repeatable way. Read-only; inspects the local Windows certificate stores only. Cisco ISE internals deferred to vendor docs. #> [CmdletBinding()] param([Parameter(Mandatory=$true)][string]$IssuingCaName)

How to run: Elevated on the device: powershell -ExecutionPolicy Bypass -File .\CISECERT-Validation.ps1 -IssuingCaName "your-issuing-ca-name" -IssuingCaName is matched against certificate Issuer/Subject to identify the VPN certificate chain. EXPECTED RESULTS PASS for: issuing CA root trusted; a leaf cert issued by that CA present in LocalMachine\My; chain valid; Client Authentication EKU present; not expired; plus the Subject/SAN printed so you can confirm it matches what Cisco ISE keys on. Any FAIL maps to Troubleshooting A/B. .NOTES White-label — the repeatable way. Read-only; inspects the local Windows certificate stores only. Cisco ISE internals deferred to vendor docs. #> [CmdletBinding()] param([Parameter(Mandatory=$true)][string]$IssuingCaName)